

Data Science

Linear Discriminant Analysis

Supervised latent analysis

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What is the lecture about?

- ★ To introduce a **supervised** definition of latent factors
- ★ To propose a conditional data modeling
- ★ To perform classification of unlabeled items

Reading: [1] (chap 4) and [2] (chap 9)

Supervised learning

Given data $\mathcal{X} \subset \Omega \subseteq \mathbb{R}^D$ associated with categories (class labels) $\mathcal{Y} = \llbracket M \rrbracket \subset \mathbb{N}$, supervised learning is about finding (learning) the parameters θ of a learner (function) ϕ_θ so as to minimize the loss $\mathcal{L}_{\mathcal{X} \times \mathcal{Y}}(\theta)$ incurred when predicting class labels using ϕ_θ

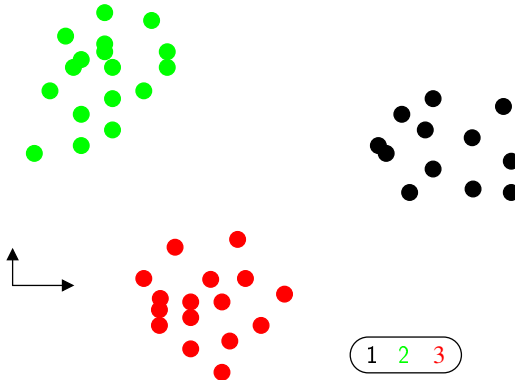
$$\begin{aligned}\phi_\theta &: \Omega \rightarrow \mathcal{Y} \\ \mathbf{x}_i &\mapsto \phi_\theta(\mathbf{x}_i) = \tilde{y}_i\end{aligned}$$

Examples

- ★ ϕ_θ is a **logistic regression** with parameters θ
- ★ ϕ_θ is a **neural network** with weights θ
- ★ $\mathcal{L}_{\mathcal{X} \times \mathcal{Y}}(\theta) = \sum_{\mathbf{x}} \|\phi_\theta(\mathbf{x}_i) - y_i\|_{\text{some}}^2$
- ★ ...

Linear Discriminant Analysis (LDA)

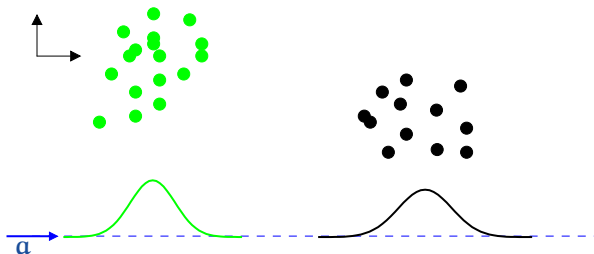
Q: Given multivariate data $\mathcal{X}$, associated with labels from $\mathcal{Y}$, we seek a model for this data



Note: This LDA is not to be mixed with “Latent Dirichlet Allocation” (related to NLP models)

Discriminant direction

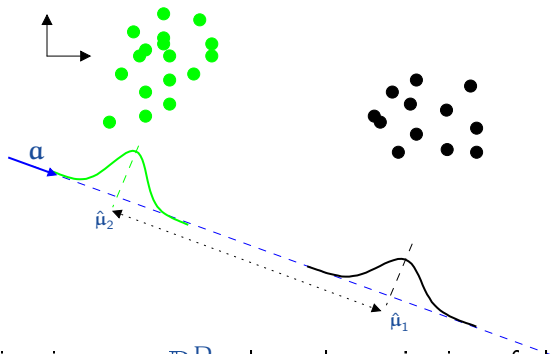
With 2 classes



we seek a direction $\mathbf{a} \in \mathbb{R}^D$ where the projection of the data over this direction $\hat{\mathcal{X}} = \text{Proj}_{\mathbf{a}}(\mathcal{X})$ shows optimal properties for class discrimination

Discriminant direction

With 2 classes



we seek a direction $\mathbf{a} \in \mathbb{R}^D$ where the projection of the data over this direction $\hat{\mathcal{X}} = \text{Proj}_{\mathbf{a}}(\mathcal{X})$ shows optimal properties for class discrimination

Inter-class discrimination criterion

If the class projections are well-separated, then they can be easily discriminated:

- ⇒ Search for direction $\mathbf{a}$ where the inter-class discrimination is maximum
- ★ Clearly, $\mathbf{a}$ must be colinear to the line $\mu_1 - \mu_2$ across the centers of mass of the classes, since

$$\|\hat{\mu}_1 - \hat{\mu}_2\|^2 = \left\| \frac{\mathbf{a}^\top \mu_1}{\|\mathbf{a}\|^2} \mathbf{a} - \frac{\mathbf{a}^\top \mu_2}{\|\mathbf{a}\|^2} \mathbf{a} \right\|^2 = \frac{1}{\|\mathbf{a}\|^2} (\mathbf{a}^\top (\mu_1 - \mu_2))^2$$

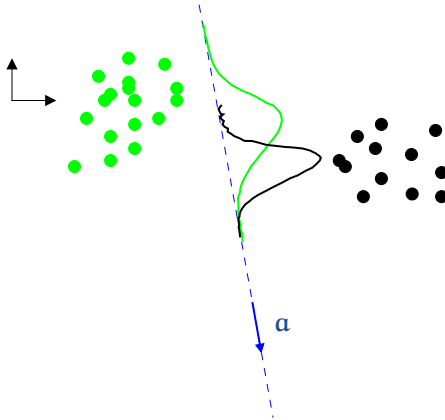
is maximum when $\mathbf{a} = \lambda(\mu_1 - \mu_2)$



- ⚠ The original data ($\mathbf{x}_i \in \mathcal{X}$) does not change. It is only the projected data ($\hat{\mathbf{x}}_i \in \hat{\mathcal{X}}$ and all subsequent computations) that changes when evolving $\mathbf{a}$

Note: ☞ prove that the center of mass of projected data is the projection of the center of mass of the original data

Intra-class discrimination criterion



We also seek to **minimize** the variance of the individual projected class

Intra-class discrimination criterion

- ★ We account for the *intra*-class variance of the projected data
- ★ Fisher criterion maximizes

$$\operatorname{argmax}_{\mathbf{a}} \frac{\|\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_2\|^2}{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}$$

where $\hat{\sigma}_k$ is the normalized variance of the projection of class k

$$\hat{\sigma}_k^2 = \frac{1}{N_k} \sum_{y_i=k} (\hat{\mathbf{x}}_i - \hat{\boldsymbol{\mu}}_k)^T (\hat{\mathbf{x}}_i - \hat{\boldsymbol{\mu}}_k)$$

Generalization: inter-class criteria

- ★ Let $\mathbf{B} = [\boldsymbol{\mu}_1 - \boldsymbol{\mu}, \dots, \boldsymbol{\mu}_M - \boldsymbol{\mu}]$ be the matrix of centered data centers ($\boldsymbol{\mu} = \frac{1}{N} \sum_N \mathbf{x}_i$ and $N = \sum_k N_k$)
- ★ $\boldsymbol{\Sigma}_b = \frac{1}{M} \mathbf{B} \mathbf{B}^T$ is the covariance matrix of class centers
- ★ We maximize over $\mathbf{a} \in \mathbb{R}^D$

$$\frac{1}{M} \sum_k (\hat{\boldsymbol{\mu}}_k - \hat{\boldsymbol{\mu}})^T (\hat{\boldsymbol{\mu}}_k - \hat{\boldsymbol{\mu}}) = \frac{1}{\|\mathbf{a}\|^2} \mathbf{a}^T \boldsymbol{\Sigma}_b \mathbf{a}$$

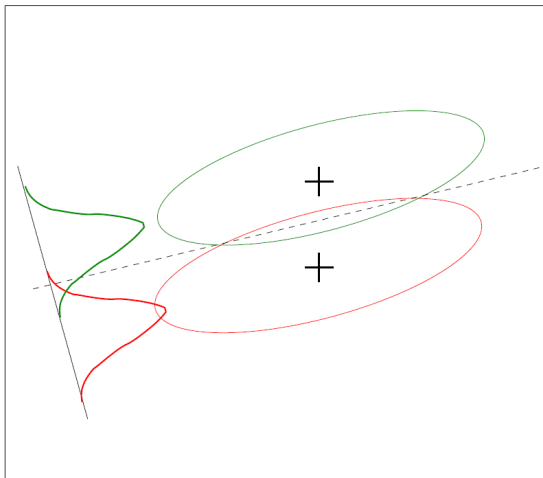
Generalization: intra-class criteria

- ★ Let $\mathbf{A}_k = [\mathbf{x}_1 - \boldsymbol{\mu}_k, \dots, \mathbf{x}_{N_k} - \boldsymbol{\mu}_k]$, $y_i = k$ be the matrix of centered data
- ★ $\frac{1}{N_k} \mathbf{A}_k \mathbf{A}_k^T$ is the *intra*-class covariance matrix
- ★ $\boldsymbol{\Sigma}_w = \sum_k \frac{1}{N_k} \mathbf{A}_k \mathbf{A}_k^T$ is the sum of *intra*-class covariance matrices
- ★ We minimize the overall distances in the projected space

$$\begin{aligned}
 & \sum_k \frac{1}{N_k} \sum_{y_i=k} (\hat{\mathbf{x}}_i - \hat{\boldsymbol{\mu}}_k)^T (\hat{\mathbf{x}}_i - \hat{\boldsymbol{\mu}}_k) \\
 &= \sum_k \frac{1}{N_k} \sum_{y_i=k} \left(\frac{\mathbf{a}^T (\mathbf{x}_i - \boldsymbol{\mu}_k)}{\|\mathbf{a}\|^2} \mathbf{a} \right)^T \frac{\mathbf{a}^T (\mathbf{x}_i - \boldsymbol{\mu}_k)}{\|\mathbf{a}\|^2} \mathbf{a} \\
 &= \sum_k \frac{1}{N_k} \frac{1}{\mathbf{a}^T \mathbf{a}} \mathbf{a}^T \mathbf{A}_k \mathbf{A}_k^T \mathbf{a} = \frac{1}{\mathbf{a}^T \mathbf{a}} \mathbf{a}^T \boldsymbol{\Sigma}_w \mathbf{a}
 \end{aligned}$$

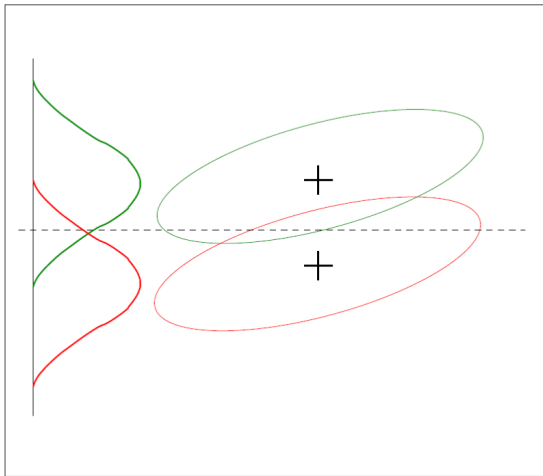
Mixing both criteria

Intra-class:



Mixing both criteria

Inter-class:



Fisher discrimination criteria

- ★ Combining intra- and inter-class criteria

$$\operatorname{argmax}_{\mathbf{a}} \mathcal{L}(\mathbf{a}) = \operatorname{argmax}_{\mathbf{a}} \frac{\mathbf{a}^T \boldsymbol{\Sigma}_b \mathbf{a}}{\mathbf{a}^T \boldsymbol{\Sigma}_w \mathbf{a}}$$

which is found if:

$$\textcircled{\Psi} \quad \frac{\partial \mathcal{L}(\mathbf{a})}{\partial \mathbf{a}} = \frac{\boldsymbol{\Sigma}_b \mathbf{a} (\mathbf{a}^T \boldsymbol{\Sigma}_w \mathbf{a}) - \boldsymbol{\Sigma}_w \mathbf{a} (\mathbf{a}^T \boldsymbol{\Sigma}_b \mathbf{a})}{(\mathbf{a}^T \boldsymbol{\Sigma}_w \mathbf{a})^2} = 0$$

$\Rightarrow$ $\mathbf{a}$ is solution of the generalized eigen system: $\boldsymbol{\Sigma}_b \mathbf{a} = \mathcal{L}(\mathbf{a}) \boldsymbol{\Sigma}_w \mathbf{a}$

Hence, $\mathbf{a}$ is the first e.vector of $\boldsymbol{\Sigma}_w^{-1} \boldsymbol{\Sigma}_b$ (with e.value $\lambda_1 = \mathcal{L}(\mathbf{a})$) $\textcircled{\Psi}$

Discriminant subspaces

- ★ eigenvectors corresponding to the largest eigenvalues λ_i are the most discriminative dimensions

$$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p \quad \text{with} \quad \lambda_1 > \lambda_2 > \dots > \lambda_p$$

- ★ M classes may be discriminated in a (at most) $(M - 1)$ -dimensional subspaces (iterative projections)

⇒ only $M - 1$ non-zero eigenvalues

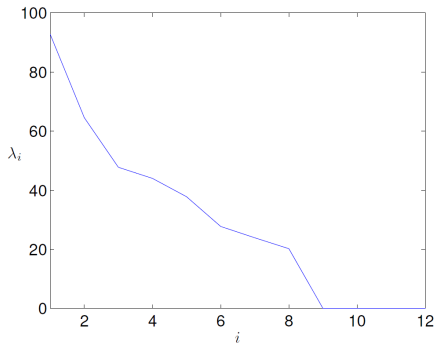
Particular case: $M = 2$

- ⊙ $\mathbf{B}\mathbf{B}^T = (\boldsymbol{\mu}_1 - \boldsymbol{\mu})(\boldsymbol{\mu}_1 - \boldsymbol{\mu})^T + (\boldsymbol{\mu}_2 - \boldsymbol{\mu})(\boldsymbol{\mu}_2 - \boldsymbol{\mu})^T \propto (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T$
- ★ hence $\mathbf{B}\mathbf{B}^T \mathbf{a}$ is a vector along direction $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$
 - ★ hence $\mathbf{a} = \lambda \boldsymbol{\Sigma}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$

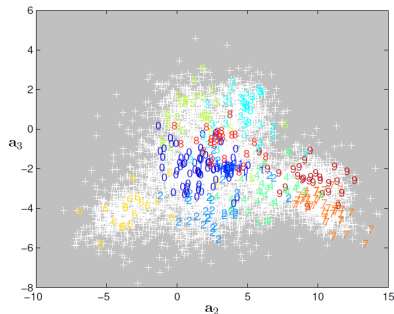
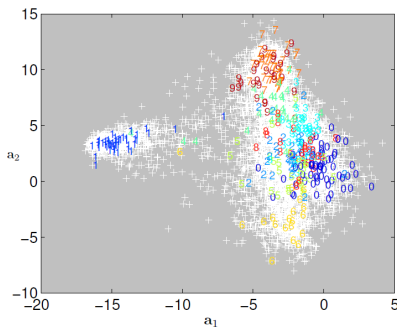
Character recognition

7291 images 16×16 (8 bits) numbers from 0 to 9

$\Rightarrow \{\mathbf{x}_i, y_i\}$ with $\mathbf{x}_i \in \mathbb{R}^{256}$ and $y_i \in \llbracket 10 \rrbracket$, $i \in \llbracket 7291 \rrbracket$



Projection



⇒ LDA finds the optimal subspace to (linearly) separate data along labels y_i .

LDA as a support for decision making

Conditional modeling

- ★ New data $j \rightarrow \mathbf{x}_j$ known, $\mathbf{y}_j$ unknown
 - ★ To which class k point j belongs? (classification)
 - ★ Declare class (categorical) random variable C
- $\Rightarrow$ Predict $\mathbb{P}(C = k | X = \mathbf{x}_j)$ (Bayes rule):

$$\mathbb{P}(C = k | \mathbf{x}_j) = \frac{\mathbb{P}(\mathbf{x}_j | C = k) \mathbb{P}(C = k)}{\mathbb{P}(\mathbf{x}_j)}$$

Gaussian approximation

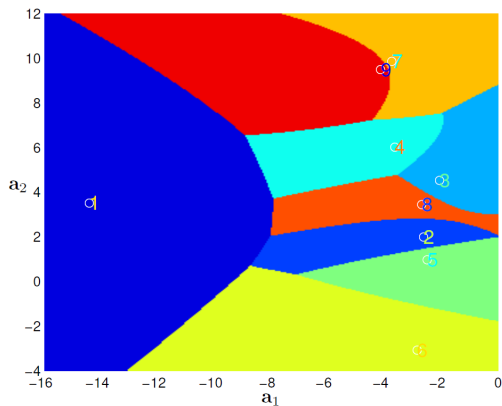
Conditional modeling

- ★ Each class is modeled by $\mathbb{P}(\mathbf{x}|\mathbf{C} = k) \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$
 - ★ Prior: $\mathbb{P}(\mathbf{C} = k) = 1/M$
 - ★ Evidence $\mathbb{P}(\mathbf{x}_j)$ is ignored
- ⇒ Maximum likelihood

$$\mathbb{P}(\mathbf{x}|\mathbf{C} = k) \simeq \exp \left(-(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \right)$$

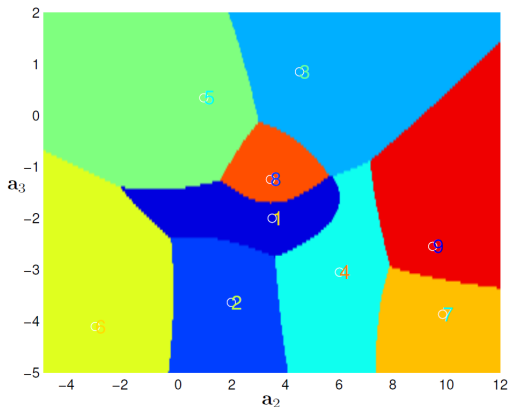
Decision (classification)

$$\delta(\mathbf{x}) = \underset{k}{\operatorname{argmax}} \mathbb{P}(\mathbf{x} | C = k)$$



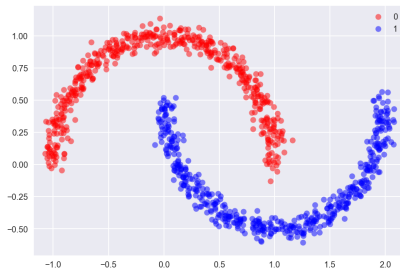
Decision (classification)

$$\delta(\mathbf{x}) = \operatorname{argmax}_k \mathbb{P}(\mathbf{x} | C = k)$$



Optimality

- ★ LDA is optimal when the M classes are each Gaussian distributed
- ⇒ because of the discrimination criteria based on covariance matrices Σ_w and Σ_b
- ★ **Linear** Discriminant Analysis → does not account for **non-linear** relationships between variables (linear discrimination)



⇒ **Non-linear** classification

Summary

- ★ LDA is a **supervised** technique
- ★ LDA finds the optimal linear latent factors explaining (linear) discrimination
- ★ Fisher linear discrimination criterion combines within- and between-scatter
- ★ LDA is resolved by finding the eigenvalues of the covariance matrix ratio ($\Sigma_w^{-1} \Sigma_b$)
- ★ These latent factors may be used for visualization (accounting for labels)
- ★ LDA (as a classification) is an example of conditional modeling: every class has a Normal distribution

Example questions [mostly require formal – mathematical – answers]

- ★ Explain the setup of supervised learning
- ★ Why is LDA a **linear** method?
- ★ What characterizes a discriminant direction? What does **α** represent?
- ★ What is the Fisher criteria?
- ★ What is the inter-class (between) criteria? Explain its principle
- ★ How to compute the inter-class covariance matrix?
- ★ What is the intra-class (within) criteria? Explain its principle
- ★ How to compute the intra-class covariance matrix?
- ★ How are they both combined?
- ★ Can you justify and explain the derivation of the maximum for **α** ?
- ★ Can you describe the **multidimensional situation** (**$D > 2$**)?
- ★ How to do inference using LDA?
- ★ Provide an example of conditional data model using LDA

References I

- [1] Christopher M. Bishop. *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Springer-Verlag, Berlin, Heidelberg, 2006. (available online).
- [2] Kevin P. Murphy. *Probabilistic Machine Learning: an Introduction*. MIT Press, 2022. (available online).

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